DELIVERABLE 5 · TESTING REPORT

What was tested, what worked, what is unverified

AI Job-Search Assistant handbook and installation package, Edition 1.0, 3 November 2026

Summary

Area	Result	Evidence
Job Finder logic (fetch, de-duplicate, filter, score, save, notify, carry-over, retry)	**Pass (37/37)**	End-to-end tests against simulated APIs (section 2)
Helper tools (chat ID, setup check)	**Pass**	Included in the 37 checks
Syntax and configuration validity (shell, Python, JSON, YAML, Compose)	**Pass**	Section 3
NAT bridge script (file-writing mode)	**Pass**	Section 4
install.sh up to the first image download	**Pass**	Section 5
Docker image downloads, model download, container build, running stack	**Not verified**	Registries were blocked in the test sandbox
Proxmox installation and host steps	**Not verified here**	No physical hardware; based on the original working build
Clean install in a fresh VM following only the handbook	**Not performed**	Recommended before publishing (section 8)
PDF output (pages, links, TOC, fonts, metadata)	**Pass**	Section 6

1. Test environment

- Isolated Linux sandbox: Ubuntu 24.04.5 LTS, 8 GB RAM, Python 3.13 (the package's Docker image uses Python 3.12), `requests` 2.34, Docker Engine 29.9.2, Docker Compose v5.6.1.
- No access to Docker Hub, GitHub Container Registry, Ubuntu/Debian mirrors, Reed, Adzuna, Telegram or Ollama model downloads from the sandbox. External services were therefore replaced by a local simulator.
- No Proxmox host or virtualisation-capable hardware was available.

2. End-to-end tests of finder.py and helpers

A local HTTP simulator imitated the Reed search and job-detail endpoints (including basic-auth checking), Adzuna search, Ollama `/api/chat` and `/api/tags`, NocoDB record endpoints and the Telegram `getMe`, `getUpdates` and `sendMessage` methods, and recorded every request. The real `finder.py` from the package was run against it with test credentials. Simulated data: 8 Reed adverts (including senior titles, a clearance requirement, an irrelevant title, a 10-day-old advert, a "Sr" title, and a full description containing a sponsorship refusal) and 2 Adzuna adverts (one duplicate of a Reed advert, one with HTML in the title and "&" in the company name).

Test	What was checked	Result
T1	Full run: Reed basic auth with key and empty password; `fullTime` / `resultsToTake=100`; Adzuna `what_phrase` / `full_time` / `max_days_old`; old advert excluded; cross-source duplicate removed; Senior/Sr/clearance/irrelevant rejected and remembered; skip phrase detected in Reed full text; exactly 3 AI calls; AI options `num_ctx 8192`, `temperature 0.1`, `num_predict 300`, `format json`, `stream false`; profile first in system prompt; description truncated; HTML stripped; 2 matches saved to NocoDB with `xc-token`, Status "Found" and the exact 8 field names; one Telegram message, best score first, HTML-escaped, correct chat ID; token never printed. (22 checks)	Pass
T2	Second run: nothing re-scored; "No new jobs above 70" message sent. (2)	Pass
T3	AI returns only whitespace: failure logged, advert NOT remembered, retried on the next run (the bug fixed in the original patch). (3)	Pass
T4	Per-run limit of 1: only the first candidate remembered; the rest carried over. (2)	Pass
T5	Placeholder Adzuna and Telegram values: Adzuna skipped and Telegram not sent, run completes. (2)	Pass
T6	Optional location: Reed `locationName` / `distanceFromLocation`, Adzuna `where` / `distance` (miles converted to km). (2)	Pass
T7	`check_setup.py --send` all OK; `get_chat_id.py` prints the chat ID; a wrong Reed key is reported with a hint and exit code 1. (3)	Pass
T8	Scheduler mode starts and stays running. (1)	Pass

Limits of these tests

The simulator follows the documented request and response shapes of each service. It cannot prove that the real services still behave the same way today, that a real 3B model returns well-formed scores, or that real NocoDB accepts the records with your table's field types. Those points were exercised by the original private build (section 7), but not by the rebuilt public code.

3. Static validation

Item	Method	Result
`install.sh`, `setup-nat-bridge.sh`	`bash -n`	Pass
`finder.py`, `get_chat_id.py`, `check_setup.py`	`python -m py_compile`	Pass
`config.json` / `answers.yaml`	JSON / YAML parsers	Pass
4 compose files (+ `extras` profile)	`docker compose config`	Pass; volume names confirmed as `ai_ollama`, `ai_openwebui`, `jobs_nocodb`

4. NAT bridge script

Run with `APPLY=0` against a copy of a typical Proxmox `/etc/network/interfaces`. It created a timestamped backup, appended a `vmbr1` block identical to the original system's configuration, wrote the forwarding sysctl file, and refused to run a second time ("vmbr1 already exists"). **Not verified:** applying it on a real Proxmox host (`ifreload -a`, live iptables rules).

5. Installer trial run

`bash install.sh -y --no-model` on the Ubuntu 24.04 sandbox:

- Pass root, OS, memory and disk checks; refusal to run on a Proxmox host is coded but could not be triggered.
- Pass base packages step (mirrors unreachable, already installed), time zone and locales.
- Pass existing Docker detected and installation skipped.
- Pass all folders and files created at the documented paths; `.env` created with mode 600; `/opt/jobagent` owned by uid 1000.
- Pass second run kept the user's `config.json`, `master.md`, `answers.yaml` and `.env`.
- Stopped at the first image download: *"failed to resolve reference … Forbidden"* (registry blocked in the sandbox). Steps after this point are **not verified**: starting Ollama/Open WebUI/NocoDB/Dockge, model download, building the Job Finder image, the placeholder check that decides whether to start it.

6. PDF checks

- Handbook: 117 A4 pages; Quick Start: 4 pages. All internal links and table-of-contents entries resolve to named destinations; page numbers were computed from the rendered PDF and confirmed stable over two passes.
- All fonts are embedded TrueType fonts; text is selectable and searchable (checked by extracting it).
- Bookmarks (outline) for every part, chapter and section.

- Every page was rendered to an image and reviewed at thumbnail size; selected pages were reviewed at full size for clipping and legibility. Code blocks wrap instead of being cut off.
- Web links in the handbook point to official project sites; they were not fetched individually from the sandbox, apart from the Proxmox download page (version 9.2 confirmed).

7. Built and used on the original system

From the project records of the original private build (16 GB laptop, Proxmox VE 9.2): Proxmox installation and networking (including the IP change after a move); private NAT bridge with port forwarding; Tailscale in the container; Docker and Dockge; Ollama and Open WebUI including the speed fix (prompt reduced from about 2,050 to about 30 tokens); n8n with a successful Telegram test message; Gotenberg health check; NocoDB tracker with the documented fields; the original Job Finder with a live first run (818 adverts fetched, 40 passed filters, scoring working) and the timeout/seen-marking patch. Also experienced there: the arm64 template mistake, the captive-portal hash errors, the USB disk dropouts and recovery.

8. Unverified items and recommended clean-install test

Unverified item	Handbook label
Rebuilt `finder.py` and helpers against the real Reed, Adzuna, Ollama, NocoDB and Telegram services	Noted in About and Ch. 20
Docker image build and full `install.sh` run with internet access	This report
Proxmox installer screens and repository steps on a fresh machine (wording may differ by version)	Illustrations marked
Lid-close setting; permanent USB autosuspend fix; Tailscale on the host	Not verified badges
4 GB, 8 GB, mini PC, existing Proxmox and VPS configurations; remote Ollama on another PC	Not verified badges
Adzuna live searches (App ID was never filled in on the original system)	Real-experience box
Backup restore procedures; TLP battery thresholds	Not verified badges
NocoDB menu wording for tokens and table ID (changes between versions)	Note box

Recommendation before public release: on any spare PC (or a Proxmox VM with nested virtualisation), follow the Quick Start from Stage D onwards using only the handbook and zip, with real keys; confirm `check_setup.py` passes, one full run produces a Telegram summary and NocoDB rows, and a reboot brings everything back. Record any wording differences and correct the handbook.